

PRODUCT REQUIREMENTS DOCUMENT

Broker Research Entitlements with Generative Search Citation Checks

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Executive Summary

This PRD outlines the implementation of entitlement checks for broker research citations within the generative search feature. The feature ensures that users only receive citations from broker research materials they are authorized to access, enforcing compliance with licensing agreements and data access policies.

1. Problem Statement

Current State

- Generative search currently surfaces citations from broker research without validating user entitlements
- Different teams and users have varying access rights to different broker research libraries
- No mechanism currently exists to filter results based on individual user entitlements
- Risk of licensing violations and unauthorized data access

Desired State

- All citations returned by generative search are filtered based on user entitlements
 - Users only see research from brokers and sources they have access rights to
 - Licensing agreements are enforced at the citation level
 - Audit trail captures citation access for compliance reporting
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2. Goals & Objectives

Primary Goals

1. Implement entitlement validation for all broker research citations in generative search results
2. Ensure compliance with licensing and data access agreements
3. Prevent unauthorized access to restricted research materials

4. Create maintainable entitlement system that scales across multiple broker libraries

Success Metrics

- 100% of citations filtered against user entitlements before display
 - Zero licensing violations related to unauthorized citation access
 - Audit logs capture 100% of citation access events
 - System latency impact less than 100ms for entitlement checks
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3. User Stories & Use Cases

Use Case 1: Licensed Research Access

Scenario: Equity analyst searches for "tech sector trends." Results include Goldman Sachs and Morgan Stanley research, both licensed for this user's team.

Expected Behavior: Both citations are displayed because the user has entitlements to both brokers.

Audit Trail: Access event logged with user ID, citation source, and timestamp.

Use Case 2: Restricted Access

Scenario: Fixed income analyst performs the same search. Results include Citi and Barclays research, but only Citi is licensed for this analyst's team.

Expected Behavior: Only Citi citation is returned. Barclays research is filtered out silently; no error is shown.

Audit Trail: System logs that Barclays citation was filtered due to missing entitlement.

Use Case 3: No Entitled Research Available

Scenario: Analyst searches for highly specialized research that returns only citations from brokers they don't have access to.

Expected Behavior: System returns "No results found" or a message indicating no entitled research is available for this query.

Audit Trail: System logs the search and the fact that all results were filtered due to entitlements.

4. Functional Requirements

FR1: Entitlement Data Model

- System must maintain a mapping of users/teams to entitled broker research sources

- Each citation must be tagged with its source broker/provider
- Entitlements must support both user-level and team-level access controls
- Support for license expiration and renewal tracking

FR2: Citation Filtering Logic

- Before returning any citation in generative search results, system must query entitlements database
- Filter out citations where user lacks entitlement to the source broker
- Return only citations user is entitled to access
- Handle edge cases: expired licenses, revoked access, invalid user states

FR3: Audit & Logging

- Log every citation access (successful return to user)
- Log every filtered citation (entitlement check failed)
- Include in logs: user ID, citation ID, source broker, entitlement status, timestamp
- Logs must be queryable for compliance and audit purposes

FR4: Entitlement Management Interface

- Admin interface to grant/revoke broker access to users and teams
- Ability to set license expiration dates
- Bulk import/export entitlements for enterprise customers
- Self-service entitlement view (users see what they have access to)

FR5: API Contract

- Entitlement check endpoint: `POST /api/v1/entitlements/check`
- Input: `user_id`, `broker_source`, `citation_id`
- Output: boolean (entitled/not_entitled) + optional denial reason
- Entitlement list endpoint: `GET /api/v1/entitlements/user/{user_id}`

5. Non-Functional Requirements

NFR1: Performance

- Entitlement check must complete in <50ms (p95)
- Caching strategy for frequently accessed entitlements
- Support for batch entitlement checks (multiple citations in one request)

NFR2: Security & Privacy

- All entitlement data encrypted at rest and in transit

- Role-based access control for admin interface (entitlements manager role)
- Audit logs are tamper-proof and immutable

NFR3: Reliability

- 99.9% uptime SLA for entitlement check service
- Graceful degradation: if entitlement service unavailable, system should default to safe mode (no citations returned)
- Automated backup and recovery of entitlements database

NFR4: Scalability

- Support for 10,000+ users and 50+ broker sources
- Horizontal scaling of entitlements service
- Ability to handle peak loads of 10,000 entitlement checks per second

6. Implementation Approach

Architecture

A dedicated entitlements microservice will be created to handle all entitlement checks and management. This service will sit between the generative search engine and the citation database, filtering results before they reach the user.

Data Storage

- **Entitlements database:** PostgreSQL with indexes on (user_id, broker_source) for fast lookups
- **Cache layer:** Redis for frequently accessed entitlements (TTL: 24 hours)
- **Audit logs:** Immutable event store (append-only log) with retention policy of 7 years

Integration Points

- **Generative search service:** Add citation source metadata to all results
- **Citation database:** Ensure all citations tagged with broker/source ID
- **User identity service:** Retrieve user team affiliation for team-level entitlements
- **Admin portal:** Web UI for entitlements management

7. Timeline & Milestones

Phase	Deliverables	Timeline
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Phase 1: Design	Architecture design, API specs, DB schema	2 weeks
Phase 2: Core Development	Entitlements service, API, admin UI	6 weeks
Phase 3: Integration	Gen search integration, testing	3 weeks
Phase 4: Testing & QA	UAT, performance testing, security audit	4 weeks
Phase 5: Launch	Staged rollout, monitoring, docs	2 weeks

Total Timeline: ~17 weeks

8. Dependencies & Risks

Dependencies

- Citation database schema must be updated to include broker/source metadata
- User identity service must provide team affiliation data
- Licensing data from contracts/procurement team

Risks & Mitigation

Risk 1: Entitlement data inconsistencies

Mitigation: Implement data validation rules and reconciliation processes

Risk 2: Performance impact on search results

Mitigation: Implement caching and batch check API; target <50ms latency

Risk 3: Service unavailability blocks search results

Mitigation: Implement graceful degradation; cache all entitlements locally

9. Success Criteria

1. All broker research citations are filtered based on user entitlements before display
2. Zero licensing violations from unauthorized citation access
3. Audit logs capture 100% of citation access events

- 4. Entitlement checks complete in <50ms (p95)
 - 5. 99.9% uptime for entitlements service
 - 6. Admin interface deployed with full entitlements management capabilities
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10. Appendix: API Specifications

Entitlement Check Endpoint

POST /api/v1/entitlements/check

Request Body:

```
1 {  
2   "user_id": "user123",  
3   "broker_source": "goldman-sachs",  
4   "citation_id": "cite456"  
5 }  
6
```

Response Body (Success):

```
1 {  
2   "entitled": true,  
3   "user_id": "user123",  
4   "broker_source": "goldman-sachs"  
5 }  
6
```

Response Body (No Entitlement):

```
1 {  
2   "entitled": false,  
3   "reason": "license_expired"  
4 }  
5
```

User Entitlements List Endpoint

GET /api/v1/entitlements/user/{user_id}

Response Body:

```
1 {  
2   "user_id": "user123",  
3   "entitled_brokers": [  
4     {  
5       "broker_id": "goldman-sachs",  
6       "expires": "2025-12-31"  
7     },  
8     {  
9       "broker_id": "morgan-stanley",  
10      "expires": "2026-06-30"  
11    }  
12  ]  
13 }  
14
```

Entitlement Management Endpoints

POST /api/v1/entitlements/grant

Grant broker access to a user or team

```
1 {  
2   "user_id": "user123",  
3   "broker_id": "goldman-sachs",  
4   "expires": "2025-12-31"  
5 }  
6
```

DELETE /api/v1/entitlements/revoke

Revoke broker access

```
1 {  
2   "user_id": "user123",  
3   "broker_id": "goldman-sachs"  
4 }  
5
```

POST /api/v1/entitlements/bulk-import

Bulk import entitlements from CSV/JSON

```
1 {  
2   "format": "csv",  
3   "data": "user_id,broker_id,expires\nuser123,goldman-sachs,2025-12-31"  
4 }  
5
```

11. Data Model

Entitlements Table

```
1 entitlements  
2 |— id (UUID)  
3 |— user_id (string)  
4 |— team_id (string, nullable)  
5 |— broker_id (string)  
6 |— granted_date (timestamp)  
7 |— expires_date (timestamp, nullable)  
8 |— status (enum: active, revoked, expired)  
9 |— created_at, updated_at (timestamps)  
10  
11 Indexes:  
12 - (user_id, broker_id)  
13 - (team_id, broker_id)  
14 - expires_date  
15
```

Audit Log Table

```
1 entitlement_audit_logs  
2 |— id (UUID)  
3 |— user_id (string)
```

```
4 |— citation_id (string)
5 |— broker_source (string)
6 |— action (enum: accessed, filtered)
7 |— reason (string, nullable)
8 |— timestamp (timestamp)
9 |— request_id (string)
10
11 Indexes:
12 - (user_id, timestamp)
13 - (broker_source, timestamp)
14 - action
15
```

12. Error Handling & Edge Cases

Expired Licenses

- System checks expiration date before allowing access
- Expired entitlements are treated as revoked
- User receives message: "Your access to this broker has expired"

Revoked Access

- Immediately blocks access to citations from that broker
- Audit log records the revocation event

Invalid User States

- User not found: treat as no entitlements
- Deactivated user: block all access
- Team membership changes: update entitlements via identity service sync

Service Degradation

- Entitlements service down: serve cached entitlements (local Redis)
 - Cache miss on unavailable service: default to safe mode (no citations returned)
 - Log all cache misses for alerting
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13. Monitoring & Alerting

Key Metrics

- Entitlement check latency (p50, p95, p99)
- Cache hit rate
- Audit log write latency
- Service error rate
- Number of filtered citations per day

- Audit log query response time

Alerts

- Entitlement service unavailable for >5 minutes
 - Check latency exceeds 100ms (p95)
 - Cache hit rate drops below 80%
 - Error rate exceeds 1%
 - Audit log write failures
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14. Security Considerations

Authentication & Authorization

- All API endpoints require valid JWT token
- Admin endpoints require "entitlements_manager" role
- User-level endpoints require user to authenticate as themselves or have admin role

Data Protection

- Entitlements data encrypted with AES-256 at rest
- TLS 1.3 for all in-transit communication
- Audit logs are immutable and append-only
- Regular automated backups with encryption

Compliance

- GDPR compliance for user data
 - SOC 2 Type II requirements
 - Audit logs retained for 7 years per regulatory requirements
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15. Future Enhancements

- Geo-based entitlements (restrict access by region)
 - Time-of-day restrictions for high-value research
 - Citation-level granular controls (entitle to specific reports)
 - Integration with external license management systems
 - Machine learning to predict entitlement changes
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Document Status: Ready for Review

Next Steps: Engineering review and technical design phase kickoff